Regression

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Regression

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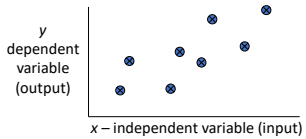
- In regression the output is continuous
  - Function Approximation
- Many models could be used – Simplest is linear regression
  - Fit data with the best hyper-plane which "goes through" the points



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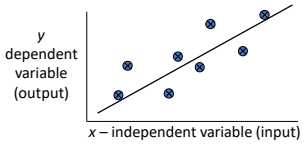

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  - Function Approximation
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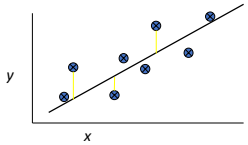
  - Fit data with the best hyper-plane which "goes through" the points
  - For each point the differences between the predicted point and the actual observation is the *residue*



Simple LR

Weight (lbs)	Height (inches)
140	60
155	62
159	67
179	70
192	71
200	72
212	75

$$Height = \beta_0 + \beta_1 \, Weight$$

$$Y = \beta_0 + \beta_1 X$$

How do we "learn" parameters

- For the 2-*d* problem (line) there are coefficients for the bias and the independent variable (*y*-intercept and slope)

$$Y = \beta_0 + \beta_1 X$$ Proof

- To find the values for the coefficients which minimize the objective function

$$\beta_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2} \qquad \beta_0 = \frac{\sum y - \beta_1 \sum x}{n}$$

Multiple LR

Price	Advertisement	Promotion	Unit Sales
$8.75	$50.04	$61.13	73959
$8.99	$50.74	$60.19	71544
$7.50	$50.14	$59.16	78587
$7.25	$50.27	$60.38	80364
$7.40	$51.25	$59.71	78771
$8.50	$50.65	$59.88	71986
$8.40	$50.87	$60.14	74885
$7.90	$50.15	$60.08	73345
$7.25	$48.24	$59.90	76659
$8.70	$50.19	$59.68	71880
$8.40	$51.11	$59.83	73598
$8.10	$51.49	$59.77	74893
$8.40	$50.10	$59.29	69003
$7.40	$49.24	$60.40	78542
$8.00	$50.04	$59.89	72543
$8.30	$49.46	$60.06	74247
$8.10	$51.62	$60.51	76253
$8.20	$49.78	$58.93	72582
$8.99	$48.60	$60.09	69022
$7.99	$49.00	$61.00	76200
$8.50	$48.00	$59.00	69701
$7.90	$54.00	$59.50	77005
$7.99	$48.70	$58.00	70987
$8.25	$50.00	$60.50	75643

X Y

$$sales = \beta_0 + \beta_1 \, price + \beta_2 \, promo + \beta_3 \, ad$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_p x_p$$

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1p} \\ 1 & X_{21} & & X_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & & X_{np} \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

$$Y = X \bullet \beta$$

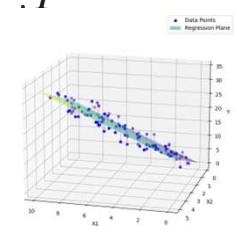
How do we "learn" parameters

- Multivariate Linear Regression

$$\hat{\beta} = (X^T . X)^{-1} . X^T . Y$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$



How do we "learn" parameters

Parameter Estimation

Least Squares

Residuals are $\epsilon = \mathbf{Y} - \mathbf{X}\beta$. Want to minimize sum of squared residuals.

$$\sum \epsilon_i^2 = [\epsilon_1 \epsilon_2 \cdots \epsilon_n] \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} = \epsilon'\epsilon$$

We want to minimize $\epsilon'\epsilon = (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$, where the "prime" $()'$ denotes the transpose of the matrix (exchange the rows and columns).
We take the derivative with respect to the vector β. This is like a quadratic function: think "$(\mathbf{Y} - \mathbf{X}\beta)^2$".
The derivative works out to 2 times the derivative of $(\mathbf{Y} - \mathbf{X}\beta)'$ with respect to β.
That is, $\frac{d}{d\beta}((\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)) = -2\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta)$. We set this equal to $\mathbf{0}$ (a vector of zeros), and solve for β.
So, $-2\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta) = \mathbf{0}$. Or, $\mathbf{X}'\mathbf{Y} = \mathbf{X}'\mathbf{X}\beta$ (the "normal" equations).

Normal Equations

$$\mathbf{X}'\mathbf{Y} = (\mathbf{X}'\mathbf{X})\beta$$

Solving this equation for β gives the least squares solution for $\mathbf{b} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$.
Multiply on the left by the inverse of the matrix $\mathbf{X}'\mathbf{X}$. (Notice that the matrix $\mathbf{X}'\mathbf{X}$ is a 2×2 square matrix for SLR.)

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

$$\hat{\beta} = (X^T.X)^{-1}.X^T.Y$$

Lasso and Ridge Regression

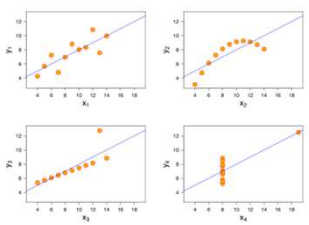
Lasso and Ridge regression are regularization techniques used to prevent overfitting in linear models by adding a penalty term to the loss function.

They improve generalization on new data by reducing model complexity, with Ridge shrinking coefficients and Lasso performing feature selection by setting some to zero

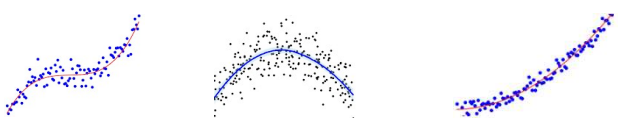
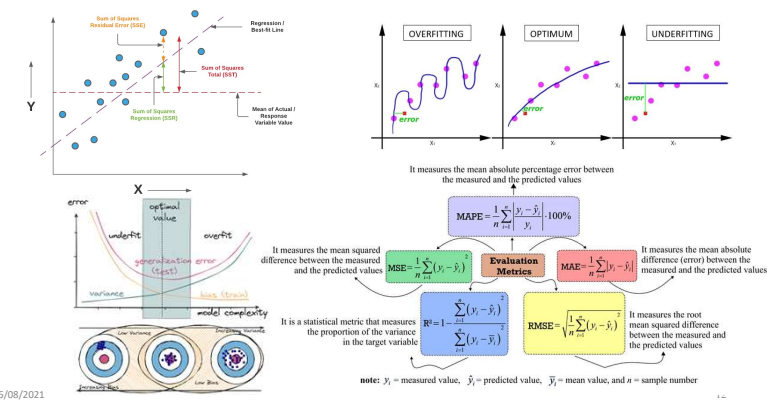
$$\text{Lasso Objective} = \sum_{i=1}^{n}(y_i - X_i\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

$$\text{Ridge Objective} = \sum_{i=1}^{n}(y_i - X_i\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

Non Linear Regression



What lines "really" best fit each case? – different approaches



Evaluation Metrics